

Solution Requirements

Date	26-june-2026
Team ID	SWUID2026223535
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

Project Design Phase-II

Solution Requirements (Functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Registration	- User enters applicant personal and financial details through the web form. - Details include name, age, income, employment, credit history, etc. - User can submit the application.
FR-2	Data Validation	- System validates all input fields for correctness and completeness. - Ensures data is in proper format and within acceptable ranges. - Displays error messages for invalid inputs.
FR-3	Credit Card Approval Prediction	- Preprocessed data is sent to the trained ML model. - Model predicts whether the credit card application will be Approved or Rejected. - Prediction is generated using the best performing model (e.g., XGBoost).
FR-4	Result Generation	- System displays the prediction result to the user. - Shows status as Approved or Rejected. - Optionally displays confidence score and rejection reasons.
FR-5	Data Storage	- Stores applicant details and prediction results in the database. - Maintains records with timestamp for each application.
FR-6	Report & History	- Allows viewing of past applications and their results. - Provides summary reports for analysis. - Supports filtering/searching by applicant details or date.

Step 2: Non-Functional Requirements (NFR)

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple, intuitive and responsive user interface for easy navigation across devices.
NFR-2	Security	User data must be protected with proper input validation, secure data transmission (HTTPS) and safe storage in the database.
NFR-3	Reliability	The system should provide accurate and consistent prediction results with minimal errors.
NFR-4	Performance	The application should generate prediction results within a few seconds for a smooth user experience.
NFR-5	Availability	The system should be available 24/7 with minimal downtime for users to access and submit applications.
NFR-6	Scalability	The system should be able to handle a large number of users and future model updates without performance degradation.